

LNS-MS SERIES

Tanker Loading Metering Skids

TECHNICAL DATA SHEET: **LNS-MS-FO**



Product Model: **LNS-MS-FO**

Application: Fuel Oil / Base Oil / Lube Oil

1. Product Overview

The **LINESIG LNS-MS-FO** Series is a modular family of skid-mounted tanker loading metering systems developed for controlled and traceable transfer of fuel oil, base oil and lube oil products. The design is intended for services where viscosity, operating temperature and pressure drop may govern meter sizing, control-valve selection and loading performance.

Design Philosophy:

Final size, flow range, pressure class, materials, meter model, heating/tracing philosophy, batch controller certification and loading interface are selected against the actual project datasheet. Coriolis mass flow measurement is preferred where technically suitable; positive-displacement metering may be evaluated where high viscosity, line size, pressure drop or project philosophy justify it.

Product Family Matrix:

Model	Primary Service	Preferred Metering	Special Design Considerations
LNS-MS-LH	Refined & light liquid hydrocarbons	Coriolis mass flowmeter	Custody/transaction measurement, loading arm interface, vapour return where required
LNS-MS-FO	Fuel oil, base oil & lube oil	Coriolis preferred; PD alternative where justified	Viscosity, heating/tracing, pressure drop, temperature compensation
LNS-MS-LPG	Liquefied petroleum gas	Coriolis mass flowmeter	Pressurised liquid service, vapour return/equalisation, ESD, anti-flash hydraulic design
LNS-MS-LNG	Liquefied natural gas	Cryogenic Coriolis mass flowmeter	Cryogenic materials, cooldown, vapour/BOG interface, ESD and thermal contraction

2. LNS-MS-FO - Fuel Oil / Base Oil / Lube Oil Loading Skid

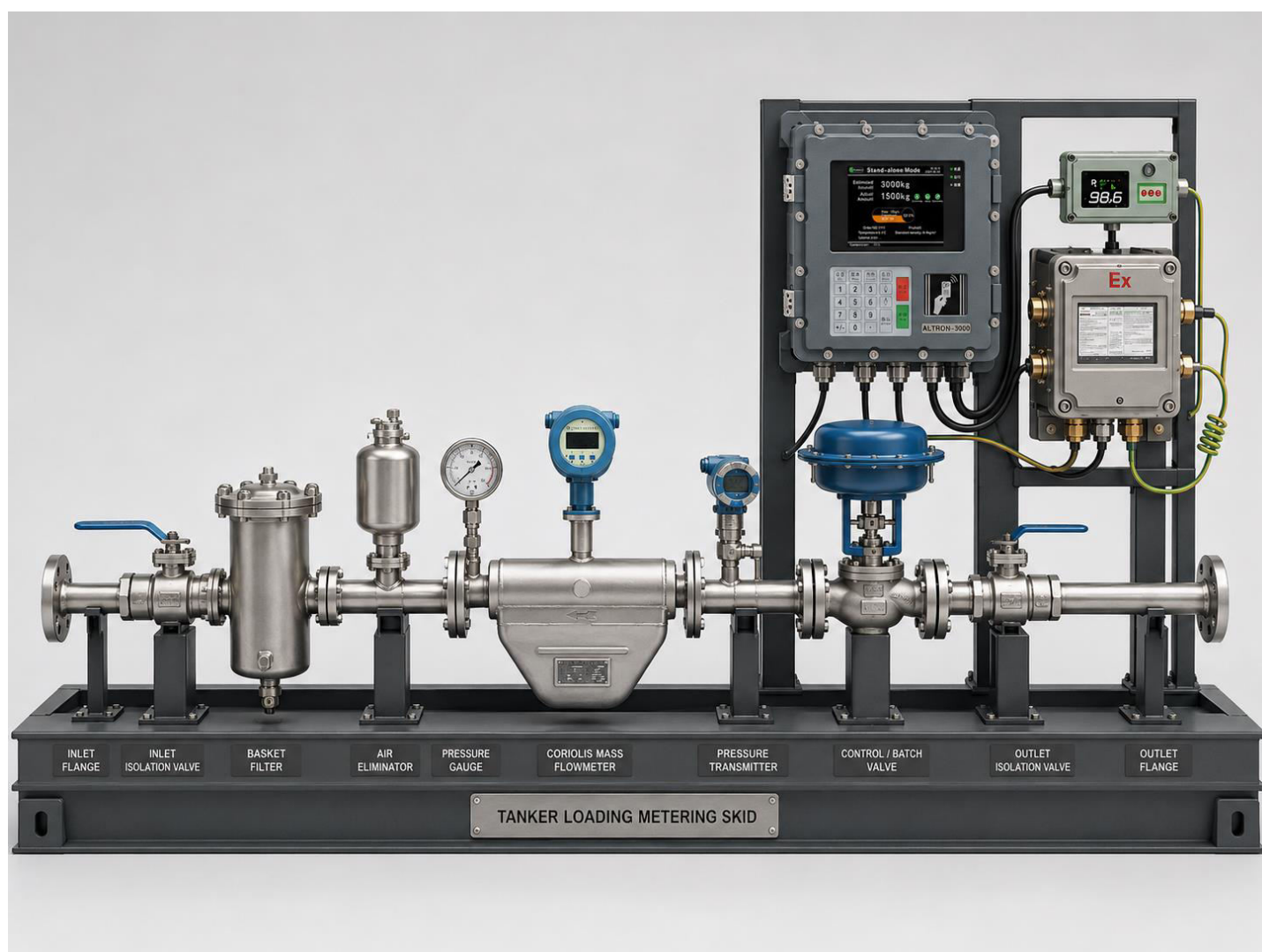
Design Aspect	LNS-MS-FO Basis
Meter	Coriolis mass flowmeter preferred; PD alternative where justified by high-viscosity/high-capacity duty
Filtration	Product-specific basket or duplex strainer/filter sized for cold-viscosity pressure drop
Air / gas elimination	Applied where entrained gas can affect liquid metering; service-dependent
Viscosity / heating	Heating, tracing, insulation or warm-up philosophy project-specific; maintain viscosity within the metering design envelope
FCV	Flow/batch control valve sized at hot and cold viscosity cases
Loading arrangement	Top, bottom or side loading according to product, tanker and terminal philosophy
Batch controller	Explosion-proof local controller or TAS-integrated controller
Safety / grounding	Grounding, overfill and ESD permissives integrated with batch logic

Standard Process Sequence:

Inlet isolation → Strainer / Filter → Air Eliminator / Gas Separator where applicable → Pressure / Temperature Measurement → Coriolis Mass Flowmeter (or approved PD alternative) → FCV / Batch Control Valve → Check / Isolation → Loading Arm or Hose Interface

Functions:

- Explosion-proof batch controller coordinates preset quantity, start/stop logic, meter totalisation and batch termination.
- Temperature status can be used as an operating permissive where viscosity control or heating is required.
- Grounding, overfill, ESD and loading-arm permissives can be hardwired into the batch logic or communicated to TAS/PLC/SCADA.
- Hydraulic sizing is checked at the highest credible viscosity and minimum operating temperature as well as at normal warm conditions.



3. General Technical Specification (LNS-MS-FO):

Parameter	Standard / Configurable Requirement
Product family	LNS-MS Series - tanker loading metering skids
Model	LNS-MS-FO - Fuel Oil / Base Oil / Lube Oil
Metering principle	Coriolis preferred where applicable; PD alternative where justified
Flow range	Project-specific; meter and line size selected from minimum / normal / maximum loading rate
Line size	Project-specific
Design pressure	Project-specific
Design temperature	Project-specific
Product properties	Density and viscosity over full operating temperature range required for final sizing
Measurement outputs	Mass flow, mass total; volumetric/density/temperature outputs where supported by selected meter
Meter accuracy	Vendor/model/project-specific; custody-transfer configuration where required
Pressure drop	Calculated for the complete skid at hot and cold viscosity cases
Process connection	ASME / EN / project flange standard
Piping material	Carbon steel / stainless steel / project-specific
Heating / tracing	Optional; selected against product viscosity and terminal utility philosophy
Hazardous area	ATEX / IECEx / NEC / local requirement as specified by project
Electrical supply	Project-specific
Instrument air	Required where pneumatic FCV or actuators are selected
Communication	Modbus RTU/TCP, Ethernet, pulse, frequency, analog/digital I/O as selected

Metering Technology Selection:

- Coriolis is preferred for direct mass measurement, strong repeatability and density/temperature diagnostics where supported.
- For high-viscosity or large-line duties, total pressure drop and meter turndown are checked before freezing the Coriolis size.
- A positive-displacement meter remains an engineering alternative only where the project duty justifies it.

Selection note:

Coriolis is preferred, not automatically mandatory. Final meter selection must account for flow range, viscosity, pressure drop, line size, custody requirements, product temperature and two-phase/flash behaviour.

4. Standard Scope of Supply

Item	Standard Scope / Description	Status
Skid base frame	Fabricated structural skid with lifting provision and equipment supports	Standard
Inlet isolation valve	Manual or actuated as specified	Standard / configurable
Strainer / filter	Product-specific filtration arrangement	Standard / configurable
DP indication	Local DP gauge and/or DP transmitter across filter where required	Recommended
Air eliminator / gas separator	Liquid-service air/gas removal where applicable	Service-dependent
Pressure gauge	Local pressure indication	Standard
Pressure transmitter	Remote pressure monitoring / batch or TAS interface	Standard / configurable
Temperature element / transmitter	Product temperature measurement	Service-dependent
Coriolis mass flowmeter	Primary metering device	Preferred standard
Flow control / batch valve	FCV, preset valve or batch shutoff arrangement	Standard
Check valve	Backflow prevention where required	Recommended
Explosion-proof batch controller	Local display, keypad, preset, totalisation and I/O	Standard for local batch control
Local ESD pushbutton	Emergency stop input to batch/terminal shutdown logic	Recommended
Grounding interface	Static-ground permissive where applicable to loading operation	Recommended
Overfill interface	Independent overfill permissive/trip where applicable	Recommended
Junction / marshalling box	Field signal and power termination	As required
Loading arm / hose interface	Mechanical outlet connection to loading system	Interface provision
Heating / tracing package	Steam, thermal-oil or electric tracing / insulation where required	Optional
Recirculation / warm-up interface	Connection or logic provision where terminal philosophy requires preheating	Optional

Explosion-Proof Batch Controller Functional Requirements:

Function	Requirement
Preset quantity	Operator/TAS entry of target batch quantity
Live indication	Flow rate, batch delivered, target, product/status
Start/stop	Local controlled batch start and stop
Valve control	Open/ramp/close FCV or batch valve according to sequence
Meter input	Mass/volume pulses, frequency, digital or communication interface
Safety permissives	Grounding, overfill, ESD, route/pump permissives as applicable
Transaction data	Batch total, start/end time, alarms and operator/order reference
Communication	PLC/TAS/SCADA integration where required
Hazardous area	Certification selected to project classification

5. Instrumentation, Control & Safety Interface:

Tag / Function	Typical Instrument / Device	Purpose
PI / PG	Pressure gauge	Local pressure indication
PT	Pressure transmitter	Remote pressure monitoring / alarms
TT / TE	Temperature transmitter / element	Product temperature / viscosity control basis
CMF	Coriolis mass flowmeter	Mass flow, batch total and supported diagnostics
FCV	Flow control / batch valve	Loading-rate control and batch termination
BC	Explosion-proof batch controller	Preset, totalisation, sequence and I/O
ESD-PB	Emergency stop pushbutton	Immediate loading stop command
GS	Static grounding monitor/interface	Ground permissive before and during transfer where applicable
OFP	Overfill protection interface	Independent overfill permissive/trip where applicable
LS/PS	Package switches as required	Interlocks / local package protection

Minimum Batch Permissive – Typical:

- Valid product / order / batch selected.
- Correct product route and transfer pump available.
- Grounding status healthy.
- Overfill protection interface healthy where provided.
- Meter and batch controller healthy.
- Product temperature within approved loading range where heating is required.
- FCV / batch valve in correct initial position.
- No active ESD or critical skid fault.

Trip / Shutdown Actions – Typical:

- ESD command: close batch/ESD valve and stop transfer-pump demand according to project cause-and-effect.
- Grounding loss during loading: controlled or immediate shutdown according to project safety philosophy where grounding is applicable.
- Overfill trip: stop batch and close transfer valve(s) where overfill protection is provided.
- Meter or controller critical fault: inhibit automatic batch completion until the approved safe fallback is defined.
- High/low pressure or abnormal process condition: alarm and trip where required by the approved design.

6. Materials, Options & Ordering Data:

Component	Typical Material / Option
Skid frame	Painted carbon steel / galvanized / stainless steel option
Process piping	Carbon steel / 304SS / 316LSS / project-specific
Meter wetted parts	Selected to product and meter model
Strainer body	CS / SS
Air eliminator	CS / SS; product-specific internals
Valves	CS / SS / project-specific
Gaskets / seals	Product and temperature compatible
Instrument tubing	316SS typical
Insulation / tracing	Project-specific where heated service is required

Available Options:

- Duplex filtration; finer product filtration; sampling point; proving/calibration connections.
- Density / temperature compensation and standard-volume calculation where required.
- Heating, insulation, tracing and recirculation / warm-up provisions.
- Additive or blend-component injection package.
- Top, bottom or side loading-arm interface.
- Printer / ticketing / RFID / vehicle identification interface.
- TAS / PLC / SCADA integration, remote I/O and historian interface.

Ordering Code – Proposed:

LNS-MS-FO / SIZE / METER / CONTROL / LOADING / OPTIONS / GROUNDING / OVERFILL / HEATING / OPTIONS

Field	Examples	Meaning
SERVICE	FO	Fuel Oil / Base Oil / Lube Oil variant
SIZE	Project DN / NPS	Nominal skid process line size
METER	CMF / PD	Coriolis mass flowmeter or approved PD alternative
CONTROL	EX-BC / TAS	Explosion-proof batch controller or TAS-integrated control
LOADING	TOP / BOTTOM / SIDE	Loading interface
GROUNDING	G	Static grounding interface
OVERFILL	O	Overfill protection
HEATING	H	Heating / tracing / insulation option
OPTIONS	A / P / R	Additive / proving / recirculation etc.

7. Project Data Required for Final Selection:

Required Input	Project Entry
Product / grade / composition	TBD
Minimum / normal / maximum flow	TBD
Batch size / tanker capacity	TBD
Design / operating pressure	TBD
Design / operating temperature	TBD
Density / viscosity versus temperature	TBD
Pour point / minimum handling temperature	TBD
Heating utility / tracing requirement	TBD
Line size / flange rating	TBD
Hazardous-area classification	TBD
Required metering accuracy / custody status	TBD
Loading method and arm/hose interface	TBD
Power supply / instrument air	TBD
Communication / TAS / PLC interface	TBD
Material / corrosion requirements	TBD

8. Document Notes:

This technical datasheet is limited to the LNS-MS-LO product configuration for Fuel Oil / Base Oil / Lube Oil. Values identified as project-specific or TBD shall be completed on the project datasheet before final equipment selection, sizing, certification or manufacture.

